

One Edge Operator

Conservation, Field Equations, and Gauge Structure from Edge Memory on the Interaction Graph

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Every quantity is quoted verbatim from execution. This paper constructs a dynamical system rather than interpreting one: a single local update rule on graph edges, from which the heat equation, the wave equation, the telegrapher equation, exact conservation, non-abelian gauge structure, and curvature are recovered as special cases or forced extensions. One target of the programme it attempts — rest mass — is not reached, and is reported as open rather than argued around.

Abstract

Conserved quantities in physics are ordinarily attributed to objects: a particle has momentum, a body has energy, a field has charge. This paper develops the alternative attribution, in which the conserved quantity belongs to the interaction rather than to either participant, and shows that under this attribution a single local update rule generates a substantial part of scalar and gauge field dynamics.

The construction places a density at each vertex of a graph and a flow on each edge, and updates the flow by one rule: the new flow is a retained fraction of the old flow minus a coupling to the density difference across the edge. The vertex plays no dynamical role; it is a compatibility constraint that redistributes density according to net edge flow. Two parameters appear, a retention coefficient and a coupling coefficient.

Conservation is automatic and exact. Because each edge contributes its flow with opposite sign to its two endpoints, the total density is conserved for every parameter choice, measured here at a drift of at most 8.9×10^{-16} over fifty steps. This is the discrete continuity equation, and it holds structurally rather than by tuning: conservation is a consequence of the edge carrying an antisymmetric flow, not a property assigned to the vertices.

The retention coefficient selects the physics. At zero retention the flow is the instantaneous gradient and the system is the heat equation, verified by monotone peak decay from 1.000 to 0.859 with no oscillation. At unit retention the flow accumulates and the system is the wave equation, verified by the signature that diffusion cannot produce: an initial central pulse splits into two counter-propagating pulses, located at positions 36 and 63 from a start at 50. Between them lies the telegrapher equation, propagation with damping. One rule, one parameter, three classical field equations.

The retention coefficient is then identified with a quantity the wider programme measures independently. It is edge memory — the degree to which a relationship carries its own past forward. Unit retention is reversible propagation with full history; zero retention is dissipation with none. This is the same axis that separates structure-building from structure-consuming dynamics in the programme's coherence measurements, now derived from the update rule rather than observed in outputs.

The construction is then pushed until it breaks. Making the edge quantity matrix-valued does not break it: it forces a parallel transporter on each edge, because matrices at different vertices cannot be compared without one, and the covariant difference that results is lattice gauge theory. Flows become non-commuting, with edge commutators reaching 2.42 against exactly zero in the abelian case, while the conserved trace is unaffected at a drift of 1.8×10^{-15} . Curvature then appears without being introduced: transport around a closed loop of four edges fails to return the identity by 0.4289 when the transporters do not commute, and by 1.3×10^{-16} when they do. Curvature is the non-commutativity of the edge connection. Rest mass is not reached, requiring a source term that survives at zero gradient, and is reported as the open target.

1 The Attribution Question

When a ball carries momentum, where is the momentum? The ordinary answer assigns it to the ball. The alternative answer, developed here, assigns it to the interaction through which it is exchanged, and treats the ball as a place where such interactions meet and must balance.

The question is not merely verbal, because the two attributions make different structural predictions. If a conserved quantity is a property of objects, conservation is a fact about how those properties happen to add. If it is a property of interactions, conservation should follow from the structure of the interaction network itself and should be unavoidable rather than coincidental.

Several physical quantities already resist the object attribution. A chemical bond energy belongs to a pair of atoms and is undefined for either alone: an isolated atom has no bond

energy, and the quantity appears only when a second atom is present at a finite distance. An electric current is a flow between points and not a property of a point. Entanglement is the extreme case, and it is not a matter of convenience: Bell's theorem establishes that no assignment of properties to the individual particles reproduces the observed correlations, so the description cannot be reduced to the participants and must be carried by the relation.

This paper takes that pattern seriously enough to build on it. If interactions are the primitive, the natural mathematical setting is a graph: vertices where interactions meet, edges that are the interactions. The question is then whether a dynamics defined on edges — with the vertices reduced to constraints — is rich enough to reproduce known physics. The paper answers for a specific and substantial class, and reports precisely where the answer stops.

2 The Construction

State. A graph carries a density at each vertex and a flow on each edge. Write the vertex density as ρ at index i , and the flow on the edge joining i and j as J . The flow is antisymmetric by construction: the flow from i to j is the negative of the flow from j to i . This antisymmetry is the entire content of the claim that the quantity belongs to the edge rather than to either endpoint, and it does all the work in Section 3.

The edge update. One rule governs the dynamics. The new flow on an edge is a retained fraction of its current flow, minus a coupling to the density difference across it:

$$J(t+1) = \alpha \cdot J(t) - \beta \cdot (\rho_j - \rho_i)$$

The retention coefficient α determines how much of the existing flow persists. The coupling coefficient β determines how strongly the flow responds to the gradient it sits on. Both are real numbers, and nothing else is specified.

The vertex constraint. Vertices do not evolve independently. A vertex density changes only by the net flow on the edges incident to it:

$$\rho_i(t+1) = \rho_i(t) - \Delta t \cdot \sum_j J_{ij}(t)$$

This is a bookkeeping identity, not a dynamical law. The vertex is a junction where edge flows must balance, and it is the discrete form of Kirchhoff's current law. All of the dynamics is in the edge update; the vertex only enforces compatibility.

3 Conservation Is Structural

Claim. *The total density is conserved for every choice of α and β .*

Argument. Each edge appears in exactly two vertex updates, contributing its flow with one sign to one endpoint and the opposite sign to the other. Summing the vertex update over all vertices, every edge contributes both plus J and minus J , and the sum telescopes to zero. The total is therefore constant in time regardless of the parameters, the initial condition, or the graph topology.

Measurement. Verified on a hundred-vertex chain over fifty steps, from an initial total of exactly 6:

α	β	Total, start	Total, end	Drift
0.0	0.20	6.000000	6.000000	0.00e+00
1.0	0.10	6.000000	6.000000	8.88e-16
0.5	0.15	6.000000	6.000000	0.00e+00

Conservation is exact to machine precision and independent of parameters. This is the point of the construction. The conserved quantity is not protected by a symmetry imposed on top of the dynamics; it is protected by the fact that a flow leaving one vertex is the same flow arriving at another. Remove the antisymmetry — allow vertex quantities to change without a corresponding edge flow — and conservation fails immediately. Conservation is a property of the edge structure.

Relation to the continuity equation. The pair of update rules is the discretisation of the continuity equation, in which the time derivative of a density is the negative divergence of a current. The density is the vertex quantity; the current is the edge quantity. The discrete statement makes visible what the continuous one obscures: the conserved total is a sum over vertices, but it is conserved because the flows live on edges and cancel in pairs.

4 One Parameter Selects the Field Equation

With conservation guaranteed for all parameters, the parameters are free to determine the physics. The retention coefficient does so, and the two limits are the two classical scalar field equations.

4.1 Zero retention: diffusion

At $\alpha = 0$ the flow retains nothing and is simply the negative gradient scaled by β . Substituting into the vertex update gives a density whose time derivative is the divergence of its gradient, which is the heat equation with diffusion constant β .

Signature. Diffusion decays a peak monotonically and never produces oscillation. Measured from an initial unit peak: 1.000 at the start, 0.957 after ten steps, 0.859 after twenty, with monotone decay confirmed at every step over the run.

4.2 Unit retention: propagation

At $\alpha = 1$ the flow accumulates rather than resetting, so the flow's increment rather than the flow itself is set by the gradient. Differentiating the vertex update once more in time gives a second time derivative equal to the divergence of the gradient, which is the wave equation with speed set by β .

Signature. The distinguishing behaviour is not decay but splitting: a wave equation takes a localised bump and separates it into two pulses travelling in opposite directions, which no diffusion can do. Measured from a bump centred at position 50 on a hundred-site chain, after sixty steps the left pulse sits at position 36 and the right pulse at position 63. The bump split and propagated.

4.3 Between them: damped propagation

For retention strictly between zero and one the system propagates like a wave and decays like a diffusion, which is the telegrapher equation. Measured at $\alpha = 0.6$, the peak falls from 1.000 to 0.778 to 0.609 while the disturbance continues to travel — the interpolation between the two limits, with no additional structure required.

Retention α	Resulting equation	Behaviour	Measured signature
0	Heat	Decay, no propagation	peak 1.000 $\rightarrow$ 0.859, monotone
$0 < \alpha < 1$	Telegrapher	Damped propagation	peak 1.000 $\rightarrow$ 0.609, travelling
1	Wave	Propagation, no decay	split to positions 36 and 63

5 Retention Is Edge Memory

The retention coefficient admits a direct physical reading, and it is the reading that connects this construction to the rest of the programme.

The coefficient measures how much of a flow persists from one step to the next — how much a relationship carries its own past forward. At unit retention the edge remembers its state completely: the dynamics is reversible, the disturbance propagates without loss, and history is preserved. At zero retention the edge remembers nothing: each step recomputes the flow from the instantaneous gradient, the dynamics is dissipative, and an arrow of time appears. Between them the edge partly remembers.

This is the same axis the wider programme measures by a different route. A coherence functional applied to evolving grids rises under reversible, structure-building dynamics and falls under irreversible, structure-consuming ones, and the sign of its change separates the two classes. That separation was an empirical observation about outputs. Here it is a parameter in the generating rule: retention near one builds and preserves, retention near zero consumes. The observed axis and the constructed parameter are the same quantity approached from opposite directions, which is the kind of agreement that gives a framework purchase.

In the programme's own terms, retention is where its two constraints meet. The requirement that every state have a successor distinct from itself forbids a frozen edge and so forbids the dynamics from stopping. The requirement that history be preserved is maximal at unit retention and absent at zero. The physics obtained — reversible waves at one end, irreversible heat at the other — is the consequence of where a system sits between those two demands.

6 Pushing Until It Breaks: Matrix-Valued Edges

A construction is worth only as much as the attempts made to break it. The natural extension, and the one the programme's own targets demand, is to make the edge quantity matrix-valued, since spin and colour charge are not scalars.

The extension is forced, not chosen. If the vertex quantities are matrices, the difference across an edge is not immediately meaningful, because the two matrices are expressed in frames attached to different vertices. To subtract them one must first transport one to the other, which requires a group element on the edge. The update becomes:

$$J(t+1) = \alpha \cdot J(t) - \beta \cdot (\rho_j - U \rho_i U^\dagger)$$

The transporter U is not an addition to the theory. It is the minimum structure required for the subtraction to be defined. What results is the covariant difference of lattice gauge theory, obtained by insisting that the original rule continue to make sense when its arguments stop being numbers.

The abelian limit is consistent. Setting every transporter to the identity, the extended construction reduces to the scalar one: the total trace is conserved with drift 8.9×10^{-16} and the edge commutators are exactly zero. The extension contains the original as a special case.

Non-trivial transporters give genuine non-abelian structure. With transporters drawn from the group generated by the Pauli matrices, the edge flows no longer commute — the commutator between neighbouring edge flows reaches 2.4198 — while the conserved trace is unaffected, drifting by 1.78×10^{-15} . Non-commutativity and conservation coexist, which is the situation in gauge theory.

Connection	Trace drift	Max edge commutator	Reading
Identity (abelian)	8.88e-16	0.00e+00	reduces to scalar case
SU(2) (non-abelian)	1.78e-15	2.4198	flows do not commute

7 Curvature Appears Without Being Introduced

Once transporters live on edges, a question arises that has no scalar counterpart: what happens to a quantity carried around a closed loop and returned to its starting vertex?

For a loop of four edges the composite transport is the product of the four transporters, and the deviation of that product from the identity is the field strength — in the language of lattice gauge theory, the plaquette. Measured with two non-commuting generators, the composite transport around the loop deviates from the identity by 0.4289. With two transporters generated by the same element, so that they commute, the deviation is 1.28×10^{-16} — zero to machine precision.

The reading. Curvature is the non-commutativity of the edge connection and nothing else. It was not added to the construction, and no geometric assumption was made. It is the observable consequence of the fact that transporting along one edge then another differs from transporting in the reverse order. A theory whose transporters all commute is flat; one whose transporters do not is curved; and the amount of curvature is the amount by which they fail to commute.

This places two of the programme's four falsification targets inside the construction rather than outside it. Spin and colour are reached by making the edge quantity matrix-valued, and curvature follows from the transporters that extension forces. Neither required a new principle.

8 What Is Not Reached

One target is not reached, and the paper does not argue around it.

Rest mass. The update rule couples flow to the density difference across an edge. Where the gradient vanishes, the coupling term vanishes, and the flow evolves only by its own retention. A rest mass is precisely a quantity that persists in the absence of any gradient — an intrinsic term that does not require a difference to exist. The construction as given has no such term, and adding one means adding a source on the edge that is independent of its endpoints, which is a genuine modification rather than a special case. Whether such a term can be motivated from the same principles, or whether it must be imported, is unresolved.

Dynamic graphs. Throughout, the graph is fixed and the transporters vary. General relativity requires the connection to react back on the geometry, so that the graph itself becomes dynamical. This construction does not do that: the plaquette measured in Section 7 is curvature on a static lattice with a given connection, not a geometry evolving under its own field equations. The extension is not attempted here.

Scope of the field-equation result. The equations recovered in Section 4 are the classical scalar field equations on a fixed graph. No claim is made about quantisation, about relativistic invariance of the discretisation, or about the continuum limit being unique. The result is that one local edge rule with one parameter generates the heat, wave, and telegrapher equations with exact conservation, which is a statement about that class and not beyond it.

9 Discussion

The construction supports a specific and limited conclusion, which is worth separating from the broader interpretation that motivated it.

What is established. A single local update rule on graph edges, with vertices acting only as compatibility constraints, conserves its total exactly for all parameters, and generates three classical field equations as parameter limits. Extended to matrix-valued quantities it forces a connection, reproduces non-abelian structure with conservation intact, and yields curvature as the non-commutativity of that connection. These are measurements, and they are reported with their numbers.

What is interpretation. The reading of the retention coefficient as edge memory, and of the two constraints of the wider programme as bounds on that coefficient, is an interpretation that the construction supports but does not prove. It is offered because it predicted the right structure — that reversibility and dissipation should be two ends of one parameter rather than

two different theories — and because it agrees with an independently measured axis in the programme's coherence work. Agreement between a constructed parameter and an observed one is evidence, not proof.

What the attribution question gains. The original question was whether conserved quantities are better attributed to objects or to interactions. The construction gives a concrete argument for the second: under the edge attribution, conservation is not an additional law but an unavoidable consequence of a flow having two ends. One cannot write down the dynamics without conservation appearing. Under the object attribution, conservation must be imposed. That asymmetry is the strongest thing this paper has to say, and it is a structural observation rather than a metaphysical one.

10 Open Edges

Rest mass, as Section 8 states: whether an intrinsic term surviving at zero gradient can be derived rather than imported.

Dynamic graphs: whether the connection can be made to react on the graph structure, which is the requirement for anything resembling general relativity.

Whether the retention coefficient is fixed by anything. It is a free parameter here, and the physics obtained depends entirely on it. Whether a system's retention is determined by its constitution, or is genuinely free, is not addressed.

Whether the continuum limit is unique. The discretisation chosen recovers the standard equations, but no argument is given that it is the only local edge rule that does so.

Whether the observed coherence axis and the constructed retention parameter are the same quantity in a stronger sense than agreement of sign. A quantitative correspondence — retention predicting the measured coherence change — is testable and untested.

Status Ledger

Claim	Status	Basis (live measurement)
Total conserved for all parameters	PROVEN	Drift $\leq 8.88\text{e-}16$ over 50 steps
$\alpha = 0$ gives the heat equation	PROVEN	Monotone decay $1.000 \rightarrow 0.859$
$\alpha = 1$ gives the wave equation	PROVEN	Pulse split, positions 36 and 63

Claim	Status	Basis (live measurement)
Intermediate α gives damped waves	PROVEN	peak 1.000 $\rightarrow$ 0.609, travelling
Matrix extension forces a connection	PROVEN	Subtraction undefined without transport
Abelian limit recovers the scalar case	PROVEN	Commutator exactly 0.00e+00
Non-abelian flows with conservation intact	PROVEN	Commutator 2.4198, drift 1.78e-15
Curvature = non-commutativity of connection	PROVEN	Plaquette 0.4289 vs 1.28e-16 abelian
Retention is the edge-memory axis	INTERPRETED	Agrees in sign with coherence axis
Rest mass from the same operator	OPEN	No term survives at zero gradient
Dynamic graph / geometry back-reaction	OPEN	Graph held fixed throughout
Retention fixed by constitution	OPEN	Free parameter here
Continuum limit unique	OPEN	One discretisation shown, not uniqueness
Quantitative retention-coherence match	OPEN	Sign agreement only; testable

Appendix: Live Output, Quoted Verbatim

Conservation, independent of parameters.

(beta only) alpha=0.0, beta=0.2: sum start=6.000000, end=6.000000, drift=0.00e+00
(alpha=1) alpha=1.0, beta=0.1: sum start=6.000000, end=6.000000, drift=8.88e-16
(mixed) alpha=0.5, beta=0.15: sum start=6.000000, end=6.000000, drift=0.00e+00

The three field equations from one parameter.

CASE A: alpha=0 => HEAT EQUATION

peak height over time: 1.000 -> 0.957 -> 0.859 (monotone decay)
monotone decay (diffusion signature): True

CASE B: alpha=1 => WAVE EQUATION

initial bump at center (50). left pulse at 36, right pulse at 63
two counter-propagating pulses (wave signature): True

CASE C: $0 < \alpha < 1$ => TELEGRAPHER (damped waves)

peak: 1.000 -> 0.778 -> 0.609

Matrix-valued extension: abelian limit and non-abelian case.

TRIVIAL connection (all U = identity):

total trace: start=4.000000, end=4.000000, drift=8.88e-16
max |[J_k, J_{k+1}] (commutator): 0.00e+00

NON-TRIVIAL connection (U = random SU(2) on each edge):

total trace: start=4.000000, end=4.000000, drift=1.78e-15
max |[J_k, J_{k+1}] (commutator): 2.4198

Curvature as the plaquette.

deviation from identity (non-commuting generators): 0.428921

ABELIAN control (same generator): 1.28e-16

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